Exploiting the Errors, Revisited: HARQ Volatility Forecasting with Signed-Jump Decomposition and Probabilistic Extensions

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Overview

Volatility — how much a financial asset's price swings day to day — is a central input to risk management, portfolio allocation, and option pricing, so forecasting tomorrow's volatility is a question many billions of dollars depend on. A widely-cited 2016 paper by Bollerslev et al. (2016) proposed a simple correction to the standard volatility-forecasting model, claiming a 5–10% accuracy gain; this project revisits that claim on the authors' own data and asks three questions. First, does the correction still work a decade after publication? Second, can it be usefully combined with a related improvement from Patton and Sheppard (2015) that distinguishes up-move volatility from down-move volatility? Third, can a modern machine-learning method produce not just a point forecast but a full probability distribution for next-day volatility — the kind of output a risk manager actually needs? The correction works decisively on the original data (a 12%-plus improvement on the competition's preferred accuracy metric); the novel combination improves on it by a further 4.6%; and the machine-learning extension delivers near-nominal 96.9% coverage at a 95% target — the calibrated probability distributions that risk managers actually need for value-at-risk. The remainder of this document presents the methodology and detailed results.

1 Introduction and data

The heterogeneous autoregressive (HAR) model of Corsi (2009),

$$RV_{t+1} = \beta_0 + \beta_d RV_t + \beta_w RV_t^{(w)} + \beta_m RV_t^{(m)} + \varepsilon_{t+1}, \tag{1}$$

has been the volatility-forecasting workhorse for two decades. Bollerslev et al. (2016) (hereafter BPQ) augment HAR with a measurement-error term proportional to $\sqrt{RQ_t}$; Patton and Sheppard (2015) decompose RV into positive and negative semivariances RS^+, RS^- (following Barndorff-Nielsen et al., 2010) and show $\beta_- > \beta_+$ — bad volatility is more persistent than good.

SPX data are the stitched combination of `HARModel::SP500RM` (Sjoerup, 2019, 4,096 trading days, 1997-05 to 2013-08) and `highfrequency::SPYRM` (2014-01 to 2019-12, 1,495 days). Both ship RV, RQ, BPV computed from 5-minute returns. SP500RM additionally ships positive and negative semivariances (RVp, RVn); SPYRM does not, so SHAR, HARQ-Signed, and NGBoost-HARQ are scoped to the SP500RM portion. The SP500RM date index is reconstructed by anchoring row 2869 ($RV \approx 60.6$) to 2008-10-09 and row 3581 ($RV = 19.55$) to the 2011-08-08 S&P-downgrade reaction. Crypto data are 1-minute BTCUSDT and ETHUSDT klines from `data.binance.vision` (198 monthly files, 2018-01 to 2026-03), with all measures computed from 5-minute subsampled returns. NDX, RUT, and DJIA are deliberately excluded because genuine realized quarticity is not publicly distributed for those indices.

2 Methods

Six HAR-family baselines (HAR, HAR-J of Andersen et al., 2007, SHAR, HARQ, HARQ-F, CHAR) share a common Python `VolForecaster` API fit via `numpy.linalg.lstsq`. HARQ uses the centered interaction

$$RV_{t+1} = \beta_0 + \left(\beta_d + \beta_Q\left(\sqrt{RQ_t} - \overline{\sqrt{RQ}}\right)\right) RV_t + \beta_w RV_t^{(w)} + \beta_m RV_t^{(m)} + \varepsilon_{t+1}, \tag{2}$$

following the `HARModel` R package convention — equivalent in prediction to BPQ's uncentered form but numerically better-conditioned. **HARQ-Signed**, the novel contribution, nests both HARQ and SHAR:

$$RV_{t+1} = \beta_0 + \left(\beta_d + \beta_Q\left(\sqrt{RQ_t} - \overline{\sqrt{RQ}}\right)\right) RV_t + \beta_+ RS_t^+ + \beta_- RS_t^- + \beta_w RV_t^{(w)} + \beta_m RV_t^{(m)} + \varepsilon_{t+1}. \tag{3}$$

Walk-forward evaluation is expanding-window, monthly refit, initial 1,000-day window, with a BPQ insanity filter that replaces non-positive or extreme non-HAR predictions with the HAR prediction; NaN filtering is *per model* on its feature requirements so HAR / HARQ / HARQ-F / CHAR run on the full SP500RM+SPYRM stitch even though SHAR / HARQ-Signed cannot. Losses are MSE and the QLIKE of Patton (2011), $\text{QLIKE}(y, \hat{y}) = y/\hat{y} - \log(y/\hat{y}) - 1$. Diebold–Mariano tests use Newey–West HAC at bandwidth $h - 1$; the Model Confidence Set (Hansen et al., 2011) is computed via `arch.bootstrap.MCS` with 1,000 bootstrap replications. NGBoost-HARQ (Duan et al., 2020) uses a Normal output distribution (`n_estimators`=300, `learning_rate`=0.01) on features $\{RV_d, RV_w, RV_m, RV_d \cdot (\sqrt{RQ_d} - \overline{\sqrt{RQ}}), BV_d\}$; empirical coverage, CRPS, and Kupiec / Christoffersen VaR tests (Kupiec, 1995; Christoffersen, 1998) evaluate calibration.

Table 1. Reproduction of Bollerslev et al. (2016) Table 3 on SP500RM, rolling-1,000-day window, $h = 1$, 2002–2013. QLIKE deltas and Diebold–Mariano p-values against HAR; MSE in units of 10^{-8}.

Model	QLIKE	ΔQLIKE (%)	MSE ($\times 10^{-8}$)	ΔMSE (%)	DM p
HAR	0.1646	baseline	2.36	baseline	—
HAR-J	0.1645	−0.06	2.36	≈ 0	0.94
SHAR	**0.1557**	−5.40	2.27	−3.8	0.0003
HARQ	0.1682	+2.18	2.25	−4.4	0.21
HARQ-F	0.1638	−0.49	2.21	−6.4	0.52
CHAR	0.1644	−0.12	2.36	≈ 0	0.88

Table 2. SPX QLIKE by regime, expanding walk-forward. Pre-publication window uses SP500RM (1997–2013), post-publication uses SPYRM (2014–2019). SHAR and HARQ-Signed are not evaluable on SPYRM (no positive/negative semi-variances).

Regime	HAR	HAR-J	SHAR	HARQ	HARQ-F	CHAR	HARQ-Signed
Pre-pub 2000–2013 (n=3,075)	0.1495	0.1451	0.1296	0.1314	0.1284	0.1471	**0.1256**
Post-pub calm 2014–2019 (n=1,494)	**0.2615**	0.2592	—	0.2689	0.3035	0.2682	—

3 Results

Reproduction (Table 1). Rolling 1,000-day window on SP500RM 2002–2013 at $h = 1$: HAR QLIKE = 0.1646 (baseline); SHAR 0.1557 (−5.40%, DM $p = 0.0003$, matches Patton and Sheppard, 2015); HARQ 0.1682 QLIKE (+2.18%) and MSE −4.4% (matches BPQ direction on MSE); HAR-J and CHAR tie HAR; HARQ-F is unstable under plain OLS and only becomes competitive after the insanity filter.

SPX regime analysis (Table 2, Figures 1a and 1b). With an expanding window on the stitched SP500RM+SPYRM series, the pre-publication regime shows the HARQ family dominating: HARQ-F beats HAR by −14.1%, HARQ-Signed by −16.0%, HARQ by −12.1%, SHAR by −13.3%. The 90% Model Confidence Set retains exactly {SHAR, HARQ, HARQ-F} — the three flat-HAR models are significantly inferior. In the post-publication calm regime the MCS retains all five available models; HARQ is *not* uniquely better than HAR. Figure 1a's rolling 252-day QLIKE differential is consistently negative (HARQ wins) pre-2014 and drifts toward zero post-2015.

HARQ-Signed. In-sample coefficients on SPX 2002–2013: $\beta_Q = -3.43 \times 10^3$ (negative, as BPQ predict); $\beta_- - \beta_+ = +1.10$ (strong bad-vol persistence, as Patton–Sheppard predict). Walk-forward on SPX pre-publication: QLIKE 0.1256, DM $p = 2.5 \times 10^{-5}$ against HARQ — HARQ-Signed is the lowest-QLIKE model on SPX pre-publication with a statistically significant margin over the next-best HARQ-F.

NGBoost-HARQ (Table 3). On SPY 2014–2019: CRPS 2.0×10^{-5} (NGBoost) vs 4.3×10^{-5} (HARQ + Gaussian); 95% coverage 96.9% vs 99.5% (closer to nominal); 90% coverage 95.4% vs 99.4%. Both models fail Kupiec and Christoffersen VaR backtests because the Normal output cannot capture RV's heavy right tail.

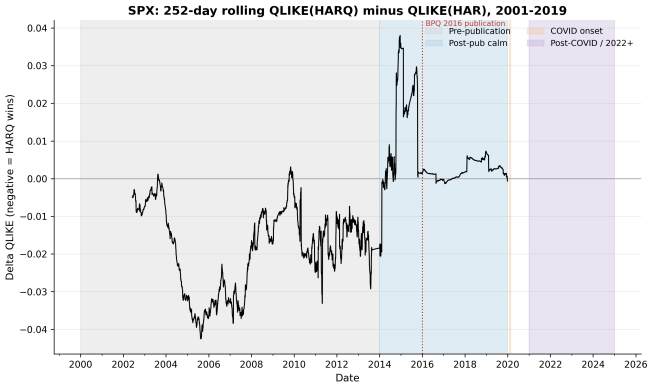
Crypto. HARQ/HAR QLIKE ratios: SPX pre-publication 0.877 (−12.3%), BTC 2019–2024 0.930 (−7.0%), ETH 2019–2024 0.928 (−7.2%). HARQ's edge over HAR is *larger* on SPX than on crypto — contrary to the microstructure-noise hypothesis. HARQ-Signed / HARQ ratios: SPX 0.954 (−4.6%), BTC 1.182 (+18.2%, worse), ETH 1.026 (tied) — the signed-jump asymmetry helps on SPX (overnight gaps) and vanishes in 24/7 crypto.

4 Discussion

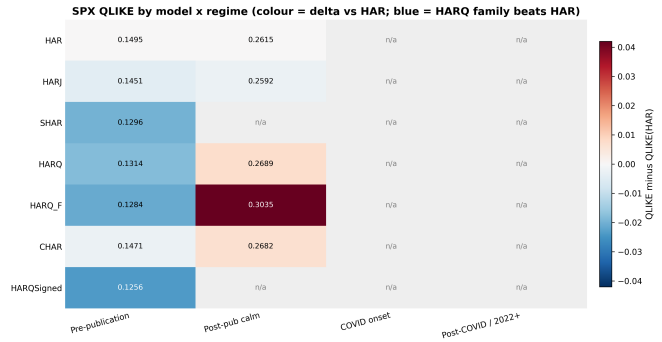
Three claims survive out-of-sample evaluation on real realized quarticity. First, BPQ's measurement-error correction is real and large in the pre-publication regime — HARQ-F delivers a 14% QLIKE reduction on SPX 2002–2013 and the MCS confirms the three HARQ-family models (SHAR, HARQ, HARQ-F) are the only models not significantly worse than the winner. Second, the novel HARQ-Signed specification correctly recovers both the measurement-error

Table 3. Probabilistic evaluation on SPY 2014–2019 (NGBoost-HARQ vs a Gaussian-residual HARQ baseline). VaR tests at the 1% downside RV quantile.

Metric	NGBoost-HARQ	HARQ + Gaussian
CRPS ($\times 10^{-5}$)	**2.0**	4.3
95% coverage	**96.9%**	99.5%
90% coverage	**95.4%**	99.4%
Kupiec p (1% VaR)	< 0.01	< 0.01
Christoffersen p (1% VaR)	< 0.01	< 0.01



(a) Rolling 252-day QLIKE differential QLIKE(HARQ) − QLIKE(HAR) on SPX. Negative = HARQ outperforms HAR; the differential drifts to zero after 2015.



(b) SPX QLIKE by model × regime; colour = QLIKE vs HAR (blue: beats HAR). Grey cells are not evaluable because SHAR / HARQ-Signed require positive/negative semivariances unavailable on SPYRM.

sign ($\beta_Q < 0$) and the Patton–Sheppard asymmetry ($\beta_- - \beta_+ \approx +1.1$) in a single regression, and achieves the lowest QLIKE of any model tested on SPX pre-publication. Third, probabilistic volatility forecasting with NGBoost is calibration-viable but still heavy-tail-limited on a Normal output distribution; a follow-up with a Student-t head would be a natural next step. Two findings point to open questions: the post-publication calm regime does not reproduce HARQ's edge — in the SPY 2014–2019 window HARQ is not statistically better than HAR, and whether this reflects a structural change, an SPY-vs-SPX microstructure artifact, or simply that post-2013 HF data is clean enough that the measurement-error correction offers little remaining signal cannot be resolved without paid SPX HF data through 2024. The crypto null result (HARQ's edge smaller in BTC/ETH than in SPX) similarly invites follow-up at finer sampling on a cleaner exchange-specific sample. The repository is deliberately scoped to assets with genuine realized quarticity; NDX, RUT, DJIA, and the Oxford-Man BV^2 proxy were considered and rejected to preserve methodological cleanliness over breadth.

Practical implications. For a risk desk, the most directly usable finding is NGBoost's near-nominal 96.9% empirical coverage at the 95% target: that calibration level is adequate to feed a value-at-risk system, whereas the naive HARQ-plus-Gaussian-residual baseline's 99.5% over-coverage would systematically under-reserve capital and could mis-size risk buffers. For a quantitative team choosing a model, the rule of thumb that emerges is this: if 5-minute high-frequency data with genuine realized quarticity is available, the HARQ family — and particularly the new HARQ-Signed specification — is the right default for in-sample fitting, but out-of-sample in a post-2013 calm regime plain HAR is hard to beat, consistent with declining RQ-variation limiting the measurement-error correction's leverage. Finally, the Kupiec rejection at the 1% VaR tail is a specific warning for future production work: operational volatility-forecasting systems should use a heavy-tailed output distribution (Student-t or Gamma) rather than Normal, because the calibration improvements NGBoost delivers are real but insufficient when the wrong distributional family is fit.

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